UNIT 2 · 9 hours

Torque Equation and Characteristics of the DC Shunt Motor



Transformers and Electrical Machines

Basic Electrical and Electronics Engineering (EEE103)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

- Derive the armature torque equation
- Distinguish armature torque from shaft torque
- Draw and explain shunt motor characteristics
- Summarise other motor types and applications

Syllabus context

DC Motor: principle of operation, back EMF and its significance, torque equation, characteristics and speed control (armature and field) of DC shunt motor.

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Deriving the torque equation

1

Power

Mechanical power = $T_a \times \omega = T_a (2\pi N/60)$.

2

Equate

$T_a (2\pi N/60) = E_b I_a$.

3

Substitute

$E_b = Z N \varphi P/(60A)$.

4

Result

$T_a = 0.159 \varphi Z I_a (P/A)$ N·m.

$T_a \propto \varphi I_a$. For a shunt motor φ is constant, so $T_a \propto I_a$.

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Armature and shaft torque

$$T_a = E_b I_a / (2\pi N/60)$$

gross (armature) torque

$$T_i = P_i / (2\pi N/60)$$

lost in iron + friction

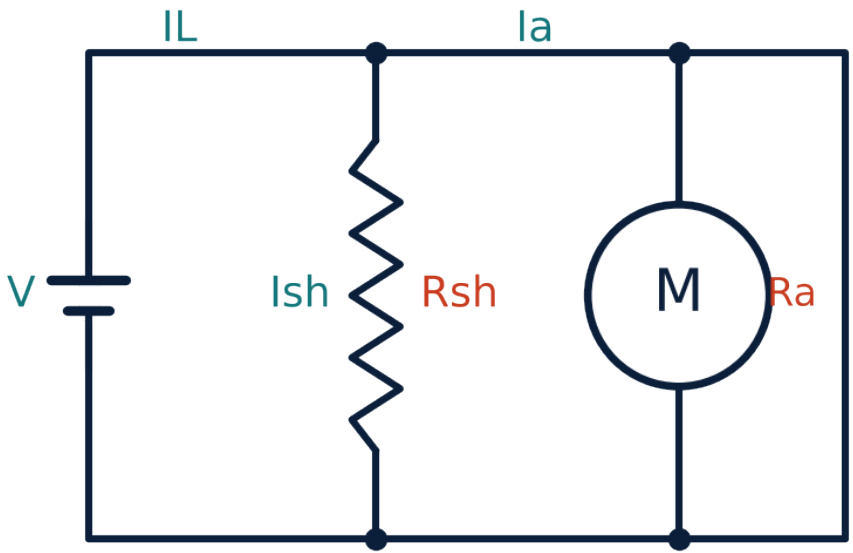
$$T_{sh} = T_a - T_i = (E_b I_a - P_i)/(2\pi N/60)$$

useful shaft torque

- Output power = Tsh × 2πN/60.
- BHP = Output / 735.5 (metric horsepower).

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Shunt motor connection



- Field winding in parallel with the armature.
- Field current Ish = V/Rsh is nearly constant, so flux is constant.
- IL = Ia + Ish.
- Eb = V − Ia Ra.

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Shunt motor characteristics

Torque vs Ia

Ta ∝ Ia: a straight line through the origin. Shaft torque is slightly lower.

Speed vs Ia

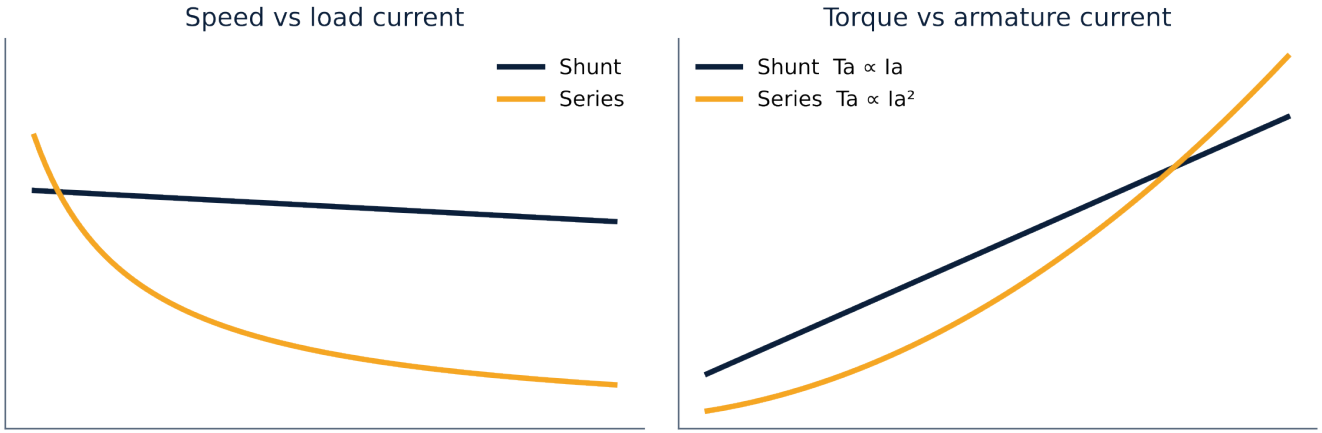
N ∝ Eb/φ. Eb falls slightly with load, so speed drops only 3–5 %: constant-speed motor.

Speed vs torque

Same shape as speed vs Ia: nearly flat, slightly drooping.

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Shunt versus series characteristics



- Shunt: speed almost constant.
- Series: $Ta \propto Ia^2$, $N \propto 1/Ia$: very high starting torque.
- Never start a series motor without load.

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Types of DC motors and applications

Type	Characteristic	Applications
Shunt	Constant speed	Lathes, pumps, fans, machine tools
Series	High starting torque	Cranes, hoists, traction
Cumulative compound	High torque, fairly steady speed	Punches, elevators, rolling mills
Differential compound	Wide speed variation	Special drives

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Numerical: series motor torque

Problem A 6-pole lap-wound 400 V series motor: Z = 920, φ = 0.045 Wb, total resistance 0.6 Ω, iron + friction loss 2 kW, line current 90 A. Find Ta, speed, Tsh and output.

- Ta = 0.159 × 0.045 × 920 × 90 × (6/6) = 592.4 N·m
- Eb = 400 − 90 × 0.6 = 346 V; N = 60 A Eb/(Z φ P) = 501.5 rpm
- Ti = 2000/(2π × 501.5/60) = 38.1 N·m; Tsh = 554.3 N·m
- Output = Tsh × ω = 554.3 × 52.5

Answer: Ta = 592.4 N·m, Tsh = 554.3 N·m, output ≈ 29.1 kW

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Key takeaways

- Ta = 0.159 φ Z Ia (P/A) N·m
- Shaft torque = armature torque − iron and friction torque
- Shunt motor: Ta ∝ Ia, almost constant speed
- Series motor: very high starting torque, never run unloaded

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Practice questions

Q1. Derive the torque equation of a DC motor.

Q2. A shunt motor has Ta = 50 N·m at Ia = 20 A. Find Ta at Ia = 30 A (φ constant).

Q3. Why is the shunt motor called a constant-speed motor?

Q4. A motor delivers 7.5 kW at 1000 rpm. Find the shaft torque.